BUILD WITH AI · CODE FOR COMMUNITIES

People's Priorities

AI for Constituency Development Planning

Turning scattered citizen requests — meetings, letters, social media, grievance portals — into a single, evidence-backed priority list an MP can act on.

- Voice
- Photo
- 3 Languages
- Live Map

build-with-ai-code-for-communities-three.vercel.app

THE PROBLEM

Citizen demand is scattered. Priorities are guesswork.

Public meetings	**Letters & calls**	**Social media**	**Grievance portals**	**Direct representations**

No objective way to consolidate feedback, spot recurring needs, or weigh competing proposals against real demand.

THE BRIEF'S OWN EXAMPLE

Requests for school upgrades vs. enrollment and travel-distance data, weighed against a competing vocational-centre proposal — which one should the MP fund first?

THE SOLUTION

One multilingual platform: submit, analyze, rank, act.

1. Citizens submit

Voice, text, or photo — in English, Hindi, or Telugu.

2. AI analyzes

Classifies category, theme, urgency & sentiment automatically.

3. Demand clusters

Recurring requests surface as hotspots on a live map.

4. MP acts

A ranked, evidence-backed priority list, ready to defend.

Works fully offline on a free rule-based NLP engine — automatically upgrades to Groq (Qwen3-32B) when a key is present. No paid services required to run or demo.

FEATURE 01

Citizen Portal

Submit a development suggestion in your own language, your own way.

3 languages

English, Hindi & Telugu, script + romanized, auto-detected.

Voice input

Web Speech API — free, no external service, works offline.

Photo evidence

Attach a picture of the pothole, leak, or building damage.

Ward tagging

Locality/ward selection grounds every request geographically.

Submit a development suggestion

Ashok Nagar ▾

"Our school roof leaks during rains, no proper drinking water — please sanction urgent repairs."

Attach a photo

Submit suggestion

FEATURE 02

AI Analysis Engine

Every submission is automatically understood — not just stored.

- Language detect
- Classify category
- Extract theme
- Score urgency
- Score sentiment

PRIMARY: GROQ (Qwen3-32B)

A single structured LLM call returns category, theme, keywords, urgency, sentiment & an English translation — for text in English, Hindi, or Telugu.

FALLBACK: rule-based multilingual engine

Zero-cost keyword matching across 3 scripts — the app works fully offline with no API key.

EXAMPLE SUBMISSION → OUTPUT

"Our school roof leaks, no drinking water — urgent, this is an emergency."

Education | Urgency 5/5 | Sentiment –0.8

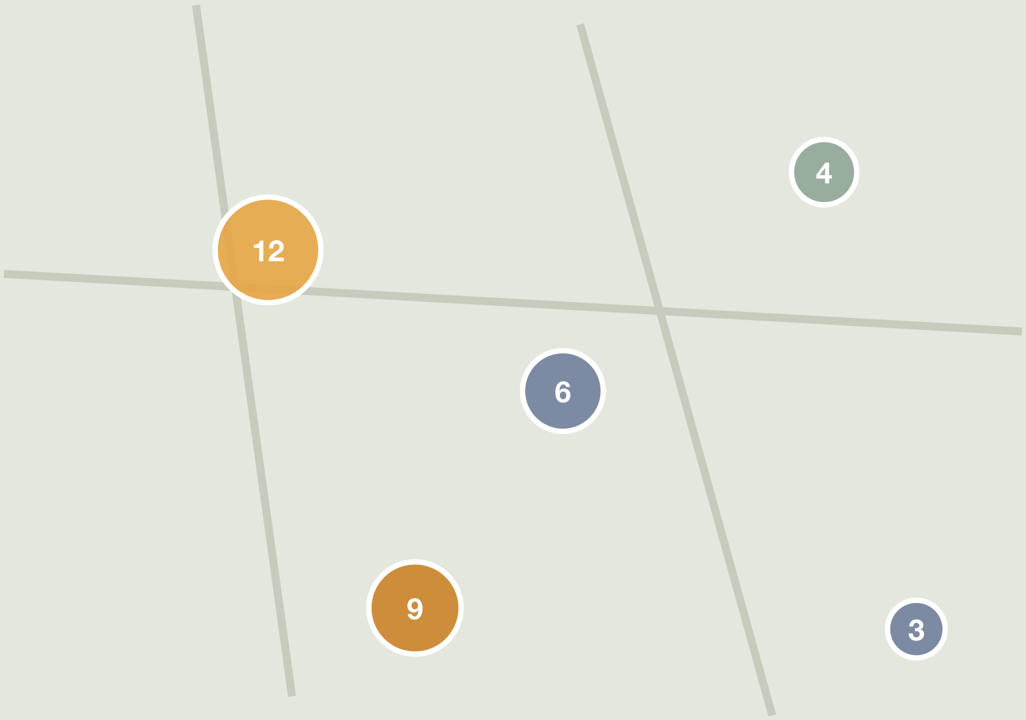
Theme: school-infrastructure-repairs

Keywords: roof-leak · drinking-water · urgent-repairs · emergency

FEATURE 03

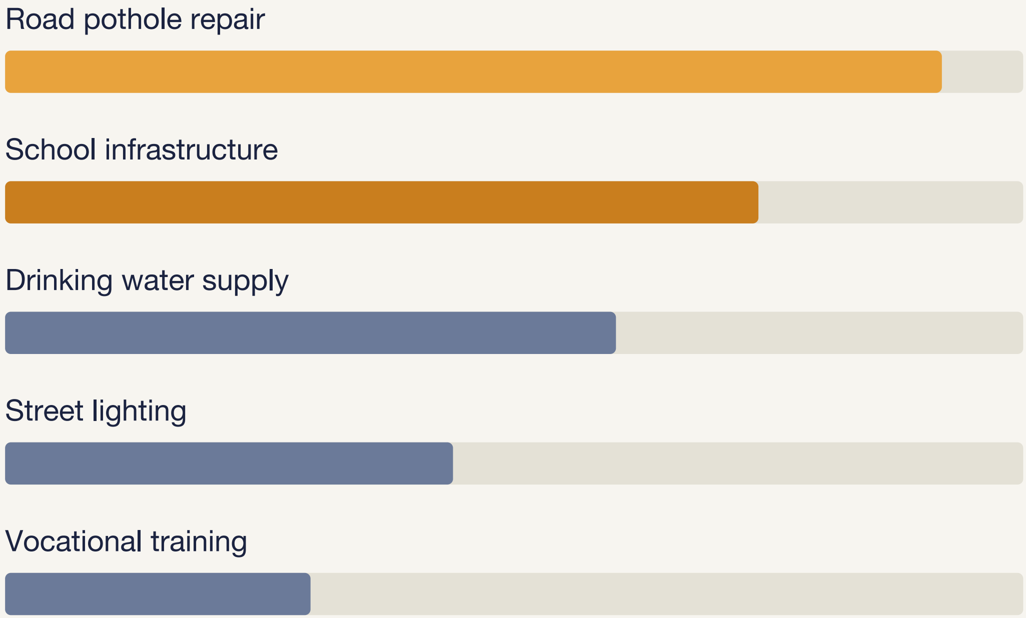
MP Dashboard: see demand before you decide

DEMAND HOTSPOTS



Circle size = submission volume · color = top category

Recurring themes, not just raw complaints



OpenStreetMap + Leaflet — zero API key, zero cost.

FEATURE 04 · THE CORE INNOVATION

A ranking engine an MP can actually defend

Four transparent, adjustable weights combine into one 0–100 score per theme-per-ward:

Demand — 35%

How many citizens raised it, relative to the largest theme.

Urgency — 25%

Average AI-derived urgency across matching submissions.

Need Gap — 30%

How badly ward demographic/infra data supports the request.

Recency — 10%

Recent surges outweigh old, possibly-resolved chatter.

WORKED EXAMPLE

A school-upgrade request in an overcrowded, long-commute ward scores higher than an identical request in a well-served ward — and can outrank a louder but less-evidenced vocational-centre proposal. Every score expands to show its breakdown and a sample citizen submission. No black box.

HOW NEED GAP IS COMPUTED

Real demographic & infrastructure data, per ward

- School enrollment & count
- Travel distance to services
- Health centre coverage
- Water access %
- Electrification %
- Road condition index
- Unemployment rate
- Literacy rate

This is what separates ranking from ranting: citizen sentiment alone can't tell you which ward is genuinely underserved — the data can.

UNDER THE HOOD

Architecture & tech stack

Citizens → **Next.js App (Portal + Dashboard)** → **Groq (Qwen3-32B) + rule-based fallback** → **Prisma ORM** → **Supabase Postgres**

Deployed on Vercel — serverless, autoscaling, zero-ops.

- Next.js 15 (App Router, TypeScript)
- Tailwind CSS
- Prisma ORM
- Supabase Postgres
- Groq · Qwen3-32B LLM
- OpenAI (secondary fallback)
- Web Speech API (voice)
- React-Leaflet + OpenStreetMap
- Recharts
- Vercel (hosting + CI/CD)

WHY THIS WINS

Built for real MP offices, not just a demo

Actually multilingual

Not translated UI strings — real script + romanized detection for Hindi & Telugu, both in NLP and voice input.

Data-backed, not opinion-backed

Ranks are computed from real demographic/infrastructure indicators, fully transparent and adjustable.

Zero-cost to run and deploy

Rule-based fallback means it works with no API keys; Groq/Supabase free tiers cover everything else.

Production-shaped, not a prototype

Postgres via Supabase, serverless deploy on Vercel, typed end-to-end with Prisma + TypeScript.

WHAT'S NEXT

Roadmap beyond the hackathon

WhatsApp & SMS intake

Meet citizens on the channels they already use — no app install.

Local dev-plan matching

Auto-match citizen themes against existing development-plan line items.

Status tracking & feedback loop

Let citizens see what happened to their request — closing the trust gap.

Open data integration

Pull live census, school, and infra datasets instead of seeded samples.

From scattered complaints to a defensible plan.

People's Priorities — AI for Constituency Development Planning

github.com/SaiBhargavRallapalli/Build-with-AI--Code-for-Communities

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